



Strathmore
UNIVERSITY

FACULTY OF INFORMATION TECHNOLOGY

BBT 2104 – INTEGRAL CALCULUS

DATE: 9TH December 2016

TIME: 2 Hours

Instruction

Answer QUESTION ONE and any other TWO questions

QUESTION ONE (30 Marks)

- a) Determine the area bounded by the graph of $y = x^2 - 5x - 6$ and the x axis. [3 Marks]
- b) Evaluate $\int \sin^5 x \, dx$ [3 Marks]
- c) Evaluate $\int \frac{(x-4)}{x(x-1)(x+2)} \, dx$ by first splitting the integrand into partial fractions [5 Marks]
- d) Integrate by parts $\int \ln x \, dx$ [3 Marks]
- e) Solve the following differential equation;
- i. $y' = \frac{2}{\sin y}$ [3 Marks]
- f) Use the method of change of variable to evaluate $\int 2x(x+2)^5 \, dx$ [4 Marks]
- g) Find the mean with respect to x of the function $f(x) = 5x^2 - 4x$ for $1 \leq x \leq 3$ [4 Marks]
- h) Find the volume of the solid of revolution formed by rotating the area enclosed by the curve $y = x + x^2$, the x axis and the points $x = 2$ and $x = 3$ through one revolution about the x axis. [5 Marks]

QUESTION TWO (20 Marks)

- a) Find the area between the curve $y = x(4 - x)$ and the x axis from $x = 0$ to $x = 5$. [5 Marks]
- b) Find the area enclosed between the curve $y^2 = 9 - x$ and the y axis [5 Marks]
- c) Evaluate the following integrals :
- i. $\int \sin^2 x \cos^5 x \, dx$ [5 Marks]

- II. Use integration by parts to evaluate $\int_1^2 x^3 \ln x \, dx$ [5 Marks]

QUESTION THREE (20 Marks)

- a) Use an appropriate method to evaluate the following integrals;
- i. $\int \cos^5 x \, dx$ [5 marks]
 - ii. $\int \cos^4 x \sin^2 x \, dx$ [6 marks]
 - iii. $\int e^x \cos x \, dx$ [6 marks]
 - iv. $\int \ln x \, dx$ [3 marks]

QUESTION FOUR (20 Marks)

- a) Evaluate using the method of change of variable [5 marks]
- $$\int x^2(2x^3 + 3)^5 dx$$
- b) Find the solution of the differential equation $xy' + 2y' - 1 = y$ giving your answer in terms of y . [5 Marks]
- c) Use Taylor's theorem to obtain a series expansion for $\cos(x + \frac{\pi}{3})$ up to and including terms in x^3 . Hence use your expansion to evaluate $\cos 62^\circ$ correct to 4 decimal places. [10 Marks]

QUESTION FIVE (20 Marks)

- a) Find the approximate value for ;
- i. $\int_0^1 e^x dx$ using the trapezium rule with 5 strips [6 Marks]
 - ii. Given $y = f(x)$ and that for given values of x , the corresponding values for y are shown in the table below, use Simpson's rule with 8 strips to approximate

$$\int_{-1}^7 f(x) dx$$

x	-1	0	1	2	3	4	5	6	7
y	17	10	5	2	1	2	5	10	17

- b) Evaluate the integral below by first expressing it as a partial fraction [6 Marks]
- [8 Marks]

$$\int \frac{5x^2 - 2x - 1}{(x + 1)(x^2 + 1)} dx$$